Have a Cow, Man (2010 Version)

Dissertation

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By

Wyatt Panaccione, B.S.

Graduate Program in Department of Nuclear Engineering

The Ohio State University

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Dissertation Committee:

Dr. Praneeth Kandlakunta, Advisor

Dr. Raymond Cao

Dr. Richard Vasques

Abstract

Field-programmable gate arrays (FPGAs) are increasingly attractive for nuclear instrumentation and control (I&C) because they offer deterministic hardware execution, reconfigurability, and high computational efficiency in compact, low-power platforms. However, their use in reactor-relevant environments is limited by radiation-induced effects such as single-event upsets and related functional disturbances. Conventional qualification campaigns can also be expensive and inefficient because they depend on scarce beam time, custom setups, and substantial external instrumentation. This work proposes a low-overhead irradiation-test methodology for commercial off-the-shelf SRAM-based FPGAs that emphasizes on-chip benchmarking, high-rate telemetry, and application-relevant fault observability.

The methodology combines two complementary workloads executed on the FPGA under identical beam conditions. The first is a flip-flop scan-farm benchmark that stresses a large population of explicitly instantiated registers and produces time-tagged mismatch events suitable for upset-rate and effective cross-section extraction. The second is a closed-loop digital control benchmark based on an I–PD controller coupled to a digital two-node thermal plant. Together, these workloads provide both fabric-level sensitivity information and application-level performance metrics, enabling comparison between raw upset behavior and functional degradation during

irradiation. Time-tagged telemetry from both workloads is streamed over UART for post-processing and comparison across test conditions.

Planned irradiations will be performed at The Ohio State University Research Reactor Fast Beam Facility, which provides a collimated fast-neutron beam with selectable gamma and thermal-neutron suppression. The intended configuration uses a bismuth gamma filter, a lead shutter, and an external cadmium filter to establish a reactor-relevant fast-neutron environment while suppressing unwanted beam components. Under the selected configuration, the facility characterization indicates a representative fast-neutron flux of approximately 2.6×10^5 cm^{-2} s^{-1} with an associated gamma dose rate of about 50 mR/h.

The digital control workload is implemented entirely on-chip using signed fixed-point arithmetic and includes a discretized two-node thermal plant, measurement lag, actuator dynamics, filtered derivative action, and anti-windup. This structure allows radiation-induced perturbations to be observed directly in control-relevant variables such as tracking error, control effort, saturation behavior, and output response. In parallel, the scan-farm benchmark repeatedly executes LOAD, HOLD, and READ phases across 55 scan chains of 512 flip-flops each, corresponding to 28,160 registers, with a PRBS31 stimulus source and embedded signature-based checking.

The goal of this work is to establish a repeatable and transferable test approach that reduces setup overhead, improves statistical confidence during limited beam allocations, and better reflects realistic FPGA workloads in nuclear environments. If successful, this combined fabric-stress and application-benchmark methodology could provide a scalable path for lower-cost comparative radiation testing of candidate FPGA platforms for reactor-relevant service.

This is dedicated to the one I love . . . la la la . . .

Acknowledgments

I thank everyone who has ever had a cow. ...

In reality, this is the only page of the dissertation which the author has full control of. You can write anything you want here, and no one can tell you it's wrong (except if the margins don't line up!!!!).

Vita

September 4, 1997	Born - Gainsville, USA
2021	B.S. Mechanical Engineering
2021-present	Graduate Teaching Associate, The Ohio State University.

Publications

Research Publications

B. Simpson "Milking a Cow". *Journal of Dairy Science*, 00(2):277–287, Feb. 1900.

Fields of Study

Major Field: Department of Nuclear Engineering

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Chapter 1: Introduction

Digital electronics increasingly occupy safety- and mission-critical roles in radiation environments. Nuclear energy systems rely on embedded computation for sensing, data acquisition, actuator control, and condition monitoring. Space and high-altitude platforms similarly depend on autonomous processing to maintain communications, navigation, and health management under particle radiation. In these settings, radiation-induced faults are not merely a reliability nuisance; they can compromise observability of the physical process, corrupt stored state, or drive a controller into an unsafe or unknown operating condition. As modern systems move toward higher integration and tighter coupling between software-defined functionality and hardware, the ability to characterize and mitigate radiation effects at the *system* level becomes increasingly important [4, 51].

Field-programmable gate arrays (FPGAs) are a compelling technology for harsh-environment instrumentation and control because they combine high-throughput digital processing with reconfigurability. A single FPGA can integrate sensor interfaces, deterministic timing, data reduction, fault detection, and control logic that would otherwise require multiple discrete components. However, many widely used FPGA families are susceptible to radiation-induced single-event effects (SEEs), including transient upsets in sequential logic and persistent upsets in configuration memory [23, 38]. These effects can manifest as straightforward bit flips in data paths, but they can also appear as subtle and difficult-to- diagnose failures such as stalled state machines, corrupted mode registers, or sporadic timing violations triggered by transients. Consequently, evaluating FPGA-based systems using only simplistic benchmarks or post-run inspection can miss the failure mechanisms that matter most in real deployments [2].

This dissertation is motivated by a practical question: *how can FPGA-based digital systems be evaluated under irradiation in a way that preserves in-situ operation, provides high observability of faults and performance degradation, and yields data products that support quantitative analysis?* Addressing this question requires more than a single measurement technique. It demands an experimental workflow that spans on-chip instrumentation, robust telemetry, test automation, and analysis methods that connect radiation events to system-level outcomes. The remainder of this chapter provides the system context for the problem, describes the radiation environments of interest, motivates the use of FPGAs while acknowledging their vulnerabilities, and outlines the in-situ test and monitoring philosophy that underpins the work.

1.1 Radiation-tolerant instrumentation and control

Digital instrumentation and control (I&C) systems form the nervous system of complex engineering facilities [35, 20]. In nuclear applications, I&C platforms acquire measurements from distributed sensors, estimate plant state, and command actuators that regulate power, temperature, and flow. Even when safety systems are implemented with conservative redundancy and independent protection channels, auxiliary digital subsystems are often responsible for data logging, operator interfaces, and supervisory control [25]. In research and test facilities, digital platforms also play a central role in experiment automation, beamline instrumentation, and synchronized acquisition of diagnostic signals.

Radiation environments challenge I&C systems in two distinct ways. The first is direct exposure to particles and secondary radiation that can produce faults in electronics. The second is the operational constraint that systems must often remain functional while data are collected. In many experiments, the most valuable information occurs *during* exposure: transient anomalies, intermittent recoveries, and performance drift with accumulated fluence. These behaviors are easy to lose when experiments rely on coarse snapshots, infrequent configuration readback, or manual intervention.

For embedded digital systems, the most consequential failures are frequently those that degrade correctness without producing an obvious halt. A sensor interface that silently drops samples, a communications link that intermittently corrupts frames, or a controller that remains stable but exhibits increased steady-state error can each undermine mission objectives while evading simple "pass/fail" criteria. Therefore, a central requirement for harsh-environment evaluation is *observability*: the ability to measure internal state, detect anomalies, and attribute system-level behaviors to underlying fault mechanisms.

In-situ observability must also be engineered within practical constraints. Telemetry bandwidth is finite, particularly when long cables, optical links, or radiation-hardened interfaces are required. Logging must tolerate resets and interruptions, and it must preserve sufficient timing context to correlate faults with external conditions such as beam transitions, actuator commands, or power supply events. These realities motivate a design philosophy in which monitoring and control are co-designed with the experiment itself.

1.2 Radiation environments and single-event effects in electronics

Radiation fields relevant to terrestrial nuclear facilities, beamline experiments, and aerospace missions vary widely in particle species and energy spectra. Fast neutrons are of particular interest for terrestrial testing because they can produce displacement damage and ionization through secondary charged particles generated in nuclear interactions with silicon and package materials [36, 62]. High-energy protons and heavy ions are often emphasized in space applications because they can directly deposit charge along a track in sensitive volumes, leading to transient and sometimes destructive events. Gamma radiation primarily contributes to total ionizing dose (TID), which can shift device parameters and increase leakage currents over time. In practical environments, mixed fields can be present, and secondary particles produced in surrounding structures can dominate the effective spectrum seen by the electronics.

Single-event effects are a family of phenomena in which a single particle interaction perturbs device behavior [11, 31]. In digital logic, the most commonly discussed SEE is the single-event upset (SEU), in which a storage element such as a flip-flop or memory bit changes state. When an upset occurs in a data register, it may appear as an incorrect numeric value that propagates through computation. When it occurs in a state register, it can change the sequencing of a controller, alter a mode selection, or prevent a process from advancing. Single-event transients (SETs) occur when a particle strike produces a short-duration pulse in combinational logic. These pulses may be filtered by logic depth and timing, or they may be captured by sequential elements depending on clock phase and pulse width [12]. More severe events such as single-event latchup (SEL) can induce a high-current state that persists until power is cycled or protective circuitry intervenes, potentially damaging the device.

A defining challenge of SEE characterization is that observed behavior depends strongly on system context. A single bit flip in a redundant arithmetic datapath may have little impact, while an identical flip in a control enable or address pointer can halt a test or invalidate measurements. Additionally, some failures are *silent*: a system may continue to run and produce outputs that appear reasonable, yet internal fault accumulation or altered timing causes gradual performance degradation. These realities motivate system-level experiments that measure not only event counts, but also the consequences of events on functional behavior [34, 24].

Quantitative analysis typically describes SEE sensitivity using an upset rate or cross-section as a function of fluence and particle spectrum [48, 53]. However, a single rate does not fully capture the engineering question of interest: *what does the system do after an event occurs?* For embedded control and monitoring platforms, the post-event trajectory—recovery, persistence, amplification, or self-correction—is often more important than the event count itself. This dissertation therefore emphasizes measurement strategies that preserve and interpret the temporal evolution of the system during irradiation.

1.3 Field-programmable gate arrays as harsh-environment digital platforms

FPGAs implement digital circuits using a fabric of programmable logic blocks, routing resources, and embedded hard macros such as block RAM, DSP slices, and high-speed serial transceivers. Their defining feature is reconfigurability: functionality is specified by a configuration bitstream that programs lookup tables, interconnect, and optional features of the device. This capability enables rapid iteration and consolidation of complex digital subsystems, which is particularly attractive in experimental environments where requirements evolve.

For instrumentation and control, FPGAs provide several practical advantages. Deterministic timing can be achieved by implementing synchronous datapaths with precisely controlled latency. Multiple heterogeneous functions—such as sensor sampling, filtering, state estimation, actuator drive, and communications—can be integrated while maintaining tight timing alignment. Hardware-level parallelism also supports high-rate data reduction and event detection, enabling long-duration experiments without saturating telemetry bandwidth.

These advantages come with a fundamental vulnerability: in many FPGA families, the configuration memory that defines the circuit is implemented using radiation-sensitive storage [23, 47]. Upsets in configuration bits can alter logic functions or routing, creating persistent errors that remain until the configuration is corrected. Upsets in user logic storage elements can corrupt state, and SETs can perturb combinational paths. Consequently, the relevant question is not whether an FPGA experiences upsets, but how the overall system is architected to *observe*, *tolerate*, and when necessary *recover* from them.

Mitigation strategies exist across multiple layers. At the circuit level, redundancy such as duplication or triplication can mask certain classes of errors, though at the cost of resource usage and potentially increased sensitivity to common-mode faults [30, 56, 43]. At the configuration level, readback and scrubbing can detect and correct configuration upsets, but they require additional infrastructure and can be difficult to integrate with strict timing requirements [18, 5, 58]. At the system level, watchdogs and health monitors can detect stalls or inconsistent outputs and trigger controlled recovery actions. The selection of mitigations is inherently tied to application constraints: a control loop may prioritize bounded recovery time, while a data logger may prioritize data integrity and provenance.

A key observation motivating this dissertation is that many experiment-relevant failure modes are *control-path* failures rather than straightforward datapath bit flips. Control paths include counters, state machines, scheduling flags, communication framing, and mode registers. Upsets in these elements can prevent valid comparisons, stop scanning, or lock a system into a non-responsive state without producing an unambiguous error signature in the primary outputs [32]. Capturing these failure modes requires on-chip observability and telemetry that are designed from the outset to detect "stuck" or "silent" conditions.

1.4 In-situ monitoring and irradiation testing methodology

In-situ irradiation testing aims to measure system behavior while the system operates under exposure, rather than treating radiation as an off-line stress applied prior to a functional check. This approach is essential when the effects of interest are transient, intermittent, or dependent on accumulated fluence. It is also essential when recovery actions—such as scrubbing, resets, or power-cycling after latchup—are part of the operational concept and therefore must be measured as part of system performance.

A practical in-situ methodology must provide fault visibility that is commensurate with the complexity of the system. For FPGA platforms, this motivates instrumentation that can detect and summarize fault activity within large collections of sequential elements and internal control structures without requiring full-state readout. The methodology must also provide time-aligned telemetry so that fault indicators can be correlated with external conditions, control commands, and measured plant response. Finally, it must support repeatable experimentation, including clear run segmentation, metadata capture, and post-processing workflows that reduce ambiguity.

One approach to fault visibility is to embed structured test logic within the FPGA design. Large sets of flip-flops can be organized into scan structures or signature-based monitors, enabling efficient detection of deviations from expected behavior. Complementary watchdog logic can supervise progress through known operational phases, flagging stalls or invalid states that might otherwise appear as a simple cessation of log messages. Telemetry formats can be designed to favor robustness and interpretability, prioritizing concise, periodic reports that survive occasional corruption and support automated parsing.

While fault detection infrastructure can quantify upset activity, system relevance is often best demonstrated using a meaningful workload. In this dissertation, closed-loop control is used as a representative workload because it couples internal computation to externally observable behavior. A controller operating on an FPGA interacts with a plant model or physical system through measured feedback. Radiation-induced faults can therefore be evaluated not only by their occurrence, but by their impact on performance metrics such as tracking error, overshoot, actuator saturation, and stability margins. Importantly, a closed- loop workload can reveal fault masking and fault amplification: some errors are corrected by feedback, while others are reinforced by the loop dynamics.

The combination of on-chip fault monitors, watchdog-based health supervision, robust telemetry, and a closed-loop workload provides a foundation for system-level irradiation studies that move beyond binary outcomes. Rather than asking whether an FPGA "survived" a run, the experiment can ask how performance evolves with exposure, which subsystems dominate observed failures, and which mitigations provide the most impact per unit resource cost.

1.5 Scope and organization of this dissertation

This dissertation focuses on radiation-induced effects in FPGA-based digital systems, with an emphasis on in-situ experimentation and system-level interpretation of faults. The work considers single-event phenomena that manifest during operation and the architectural choices that influence detectability, tolerance, and recovery. While total ionizing dose and displacement damage can influence long-term behavior, the primary emphasis is placed on single-event-driven disruptions relevant to real-time operation.

Chapter 2 provides background on the physics of radiation interactions relevant to digital electronics and reviews prior work in FPGA radiation testing and mitigation. Chapter 3 describes the hardware and software infrastructure used to conduct in-situ experiments, including embedded monitors and telemetry workflows. Chapter 4 presents the closed-loop FPGA workload used to connect internal fault activity to externally measurable performance. Chapter 5 presents irradiation results and analysis, including event classification and quantitative metrics versus exposure. Chapter 6 concludes the dissertation with a synthesis of findings and directions for future work in harsh-environment digital platforms and radiation-aware system design.

Chapter 2: Background and Prior Work

This chapter provides the technical foundation for the experimental and analytical work presented in later chapters. Section 2.1 reviews the physics of radiation interactions relevant to semiconductor devices, including the mechanisms by which energetic particles deposit charge and produce faults. Section 2.2 classifies the principal single-event effects observed in digital circuits and describes their signatures. Section 2.3 presents the architecture of field-programmable gate arrays with particular attention to the SRAM-based configuration fabric and the mapping of digital circuits onto programmable resources. Section 2.4 examines how radiation effects manifest specifically in FPGAs, distinguishing configuration memory upsets from user logic upsets and combinational transients. Section 2.5 surveys mitigation and hardening strategies at the circuit, configuration, and system levels. Section 2.6 reviews prior experimental methodologies for FPGA irradiation testing and identifies gaps that motivate the present work.

2.1 Radiation interactions with semiconductor devices

The radiation environments encountered by digital electronics range from the high-energy galactic cosmic ray spectrum in deep space to the mixed neutron and gamma fields present in terrestrial nuclear facilities [51, 62, 39]. Although the particle species, energies, and fluxes differ substantially across these environments, the fundamental mechanisms by which radiation perturbs semiconductor devices share a common physical basis: the deposition of energy in or near sensitive volumes of the device, leading to the generation of free charge carriers that can alter circuit state [3]. The historical development of single-event effects as a recognized discipline, the early observations of on-orbit anomalies in the late 1970s, and the subsequent maturation of rate prediction methodologies have been chronicled by Petersen et al. [40].

2.1.1 Ionization and charge generation

When an energetic charged particle traverses a semiconductor, it loses energy primarily through Coulomb interactions with bound electrons, liberating electron–hole pairs along its track. The rate of energy loss per unit path length in a material is described by the linear energy transfer (LET), conventionally expressed in units of $\mathrm{MeV\,cm^2\,mg^{-1}}$ to normalize for material density. In silicon, the average energy required to generate one electron–hole pair is approximately 3.6 eV [11]. A heavy ion with an LET of 10 $\mathrm{MeV\,cm^2\,mg^{-1}}$ therefore generates roughly 6.5×10^4 electron–hole pairs per micrometer of track length.

The resulting charge cloud is initially concentrated along the particle trajectory but evolves rapidly through drift under any electric field present and through carrier diffusion. In the depletion region of a reverse-biased junction, the internal electric field efficiently separates the generated carriers, producing a transient current pulse at the device terminal [10]. Outside the depletion region, charge collection occurs more slowly through diffusion, and the effective collection volume can extend well beyond the geometrical depletion width through a mechanism historically termed the funneling effect. Figures 2.1 and 2.2 illustrate this process through simulated electron concentration and electrostatic potential contours, respectively, showing how the ion track distorts both the carrier distribution and the potential landscape well below the nominal depletion boundary.

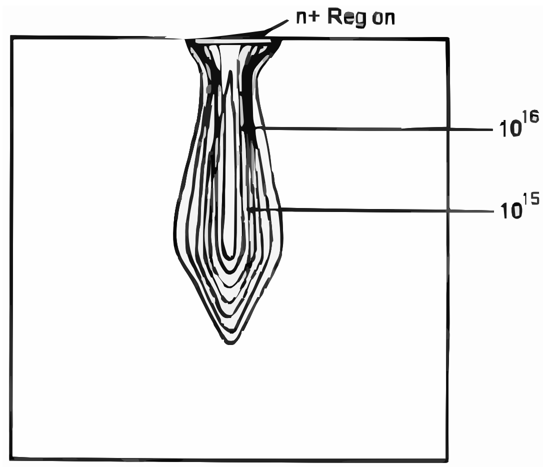


Figure 2.1: Simulated electron concentration contours following a heavy-ion strike at the drain of an n^+/p junction. Carriers generated along the ion track are swept toward the n^+ region by both the electric field and the concentration gradient, illustrating the charge-funneling mechanism. Contour levels span from 10^{15} to $10^{16}\,\mathrm{cm}^{-3}$.

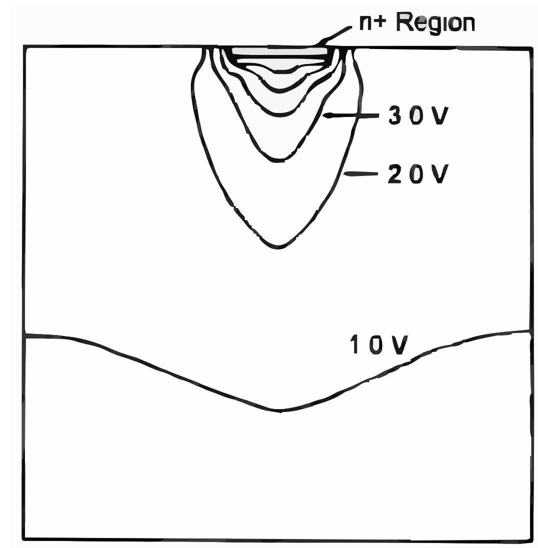


Figure 2.2: Simulated electrostatic potential contours in the same structure during charge collection. The distortion of the 1.0 V, 2.0 V, and 3.0 V equipotential surfaces along the ion track shows how the field-enhanced funneling column extends deep into the substrate, driving efficient carrier collection well beyond the nominal depletion boundary.

The magnitude and duration of the collected current pulse depend on the particle LET, the angle of incidence, the geometry of the sensitive junction, the local bias conditions, and the carrier dynamics in the specific technology node. As feature sizes decrease and supply voltages are reduced, the critical charge (Q_{crit}) required to change the state of a storage element also decreases, increasing sensitivity to lower-LET particles [4].

2.1.2 Direct ionization by heavy ions and protons

Heavy ions—nuclei heavier than helium—are the primary concern for direct ionization effects in space environments. These particles can have LETs ranging from below 1 $\mathrm{MeV\,cm^2\,mg^{-1}}$ for light ions to over 80 $\mathrm{MeV\,cm^2\,mg^{-1}}$ for very heavy species such as iron or nickel. The galactic cosmic ray spectrum contains a broad distribution of ion species with a peak flux at energies of hundreds of MeV to several GeV per nucleon.

Protons are the most abundant particle in the space radiation environment but have a very low direct LET in silicon, typically below 0.01 $\mathrm{MeV\,cm^2\,mg^{-1}}$. Nevertheless, protons with energies above approximately 10–20 MeV can produce upsets through indirect mechanisms: nuclear reactions between the incident proton and silicon or other nuclei in the device produce secondary ions with much higher LET than the primary particle. These nuclear reaction products are short-ranged but can deposit substantial charge in a small volume near the interaction site [48].

2.1.3 Neutron interactions

Neutrons are uncharged and do not directly ionize the semiconductor lattice. However, neutron–nucleus interactions can produce charged secondary particles—recoiling nuclei, alpha particles, and other fragments—that deposit energy through the mechanisms described above. In terrestrial environments, atmospheric neutrons generated by cosmic ray interactions in the upper atmosphere represent the dominant source of single-event effects at ground level [36]. The terrestrial neutron spectrum is broad, spanning thermal energies up to several GeV, with a characteristic peak near 100 MeV.

In nuclear facilities, fast neutrons from fission and spallation sources can have substantially higher fluxes than the atmospheric background. The threshold neutron energy for producing significant upsets in silicon devices depends on the technology node and the specific nuclear reaction channels accessible. Neutron-induced upsets share many characteristics with proton-induced upsets in that the actual charge deposition is performed by secondary ions rather than the primary particle [62].

2.1.4 Total ionizing dose and displacement damage

In addition to single-event phenomena, cumulative radiation exposure produces two classes of long-term degradation. Total ionizing dose (TID) arises from the gradual accumulation of charge trapped in insulating layers, particularly gate oxides and isolation structures. TID can increase leakage currents, shift threshold voltages, and reduce transistor drive strength. The effects are cumulative and largely independent of the species delivering the dose, depending primarily on the total absorbed dose in the oxide [37, 51, 31]. Fleetwood provides a comprehensive review of TID mechanisms in MOS and linear-bipolar technologies, including the important phenomenon of enhanced low-dose-rate sensitivity [15]. As feature sizes continue to shrink, the interplay between TID and single-event phenomena grows more complex, a challenge underscored by recent analyses of radiation effects in post-Moore technologies [16].

Displacement damage results from non-ionizing energy loss in which incident particles displace atoms from their lattice positions, creating vacancy–interstitial pairs and stable defect clusters. Displacement damage primarily degrades minority carrier lifetime and is most significant in bipolar technologies and optical devices. In CMOS circuits operating at moderate dose rates, TID effects typically dominate over displacement damage.

While this dissertation focuses on single-event effects that manifest during real-time operation, it is important to acknowledge that TID and displacement damage can alter the baseline operating point of a device and thereby modify its sensitivity to subsequent single events. This coupling between cumulative and acute radiation effects can be significant in long-duration experiments.

2.2 Classification of single-event effects

Single-event effects encompass a family of phenomena, all originating from the charge deposited by a single particle interaction but differing in their manifestation and severity. Figure 2.3 provides a top-level taxonomy that organizes these effects by physical origin, reversibility, and destructive potential. This section describes the principal SEE types relevant to digital logic and storage circuits.

2.2.1 Single-event upset

A single-event upset (SEU) occurs when charge deposited by a particle strike changes the stored state of a bistable element such as a flip-flop or SRAM cell [11]. The classic mechanism involves the collection of charge at the drain of an off-state transistor in a cross-coupled inverter pair. If the collected charge exceeds the critical charge of the cell, the regenerative feedback of the latch reinforces the perturbation and the cell transitions to the complementary state.

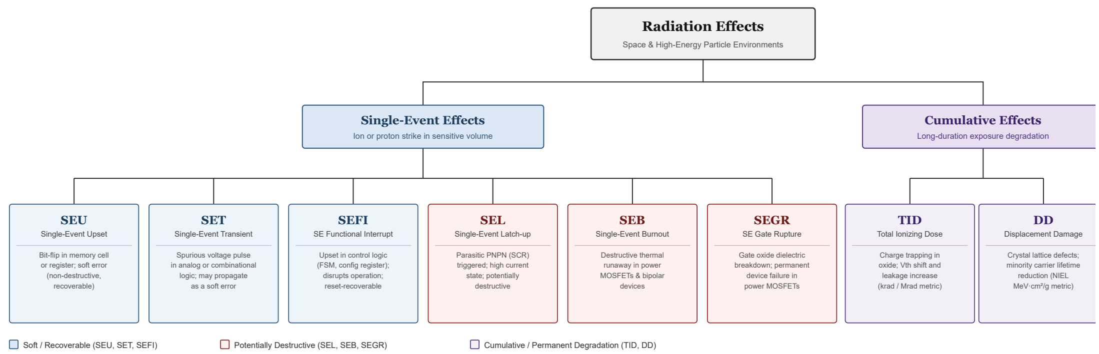


Figure 2.3: Taxonomy of radiation-induced effects in semiconductor devices. Single-event effects are subdivided into soft and recoverable effects—single-event upset (SEU), single-event transient (SET), and single-event functional interrupt (SEFI)—and potentially destructive effects—single-event latchup (SEL), single-event burnout (SEB), and single-event gate rupture (SEGR). Cumulative effects comprise total ionizing dose (TID) and displacement damage (DD), each with distinct physical mechanisms and metrology. Color coding indicates severity: blue for soft/recoverable, red for potentially destructive, and purple for cumulative/permanent degradation.

The critical charge depends on the cell capacitance, supply voltage, and feedback strength. In a standard six-transistor (6T) SRAM cell, the critical charge is proportional to $C_{\text{node}} \times V_{DD}$, where C_{node} is the total capacitance at the sensitive node and V_{DD} is the supply voltage. As technology scales and supply voltages decrease, the critical charge decreases, generally increasing SEU susceptibility [4].

SEUs in data storage elements (registers, block RAM) typically corrupt the numerical value held in the element. These upsets are often detectable through parity, error correcting codes, or comparison with redundant copies. SEUs in control-path elements (state machine registers, address counters, mode bits) can have disproportionate impact because they alter the sequencing or routing of information through the system rather than simply changing a data value.

Multiple-cell upsets (MCUs) occur when a single particle strike upsets more than one adjacent storage element. As transistor geometries shrink and cells are packed more tightly, MCUs become increasingly common and can defeat single-bit error correction schemes unless the memory is designed with interleaving or other layout strategies [11, 6].

2.2.2 Single-event transient

A single-event transient (SET) is a voltage or current pulse produced by charge collection at a node in combinational logic rather than in a storage element [12]. Ferlet-Cavrois, Massengill, and Gouker provide an extensive review of SET generation, propagation, and filtering in digital CMOS, including the dependence on technology scaling [14]. The resulting pulse propagates through subsequent logic stages and may be captured by a downstream flip-flop if it arrives during the clock sampling window with sufficient amplitude and duration. Whether a SET ultimately causes a functional error depends on three filtering mechanisms: logical masking (the pulse arrives at an input that does not affect the output), electrical masking (the pulse is attenuated by the RC characteristics of the logic gates), and temporal masking (the pulse does not coincide with the clock edge).

As clock frequencies increase and logic stages become faster, the temporal and electrical filtering of SETs becomes less effective, increasing the probability that a combinational transient is captured as a persistent error [12]. In FPGA implementations, SETs are particularly relevant in high-speed datapaths and in signal-processing pipelines where many combinational levels may exist between register stages.

Figure 2.4 illustrates the capture of an SET in a master–slave D flip-flop. The particle strike induces a current pulse whose amplitude and duration determine whether the transient crosses the logic threshold during the clock sampling window and is latched as a persistent upset. Figure 2.5 demonstrates how charge sharing between adjacent transistors can reduce the effective charge collected at the struck node, narrowing the resulting SET pulse and in some cases preventing capture altogether.

2.2.3 Single-event latchup

Single-event latchup (SEL) is a potentially destructive condition in which a particle strike triggers a parasitic thyristor (PNPN) structure inherent in bulk CMOS technology [11]. The resulting low-impedance path between the supply rails sustains a high-current state through regenerative feedback. If not interrupted by power removal or current limiting, latchup can cause permanent damage through thermal destruction of metallization or junctions.

SEL susceptibility depends on the substrate and well structures of the technology, the supply voltage, and the device temperature. Elevated temperature generally decreases the holding current and increases susceptibility. Modern CMOS technologies employ guard rings, retrograde wells, and epitaxial substrates to increase SEL immunity, but the phenomenon remains a concern for commercial-grade devices operated in radiation environments [51].

From an experimental perspective, SEL detection requires monitoring of supply current. A sustained current increase above a defined threshold, especially one that does not self-recover, is the primary indicator. Automated latchup detection and

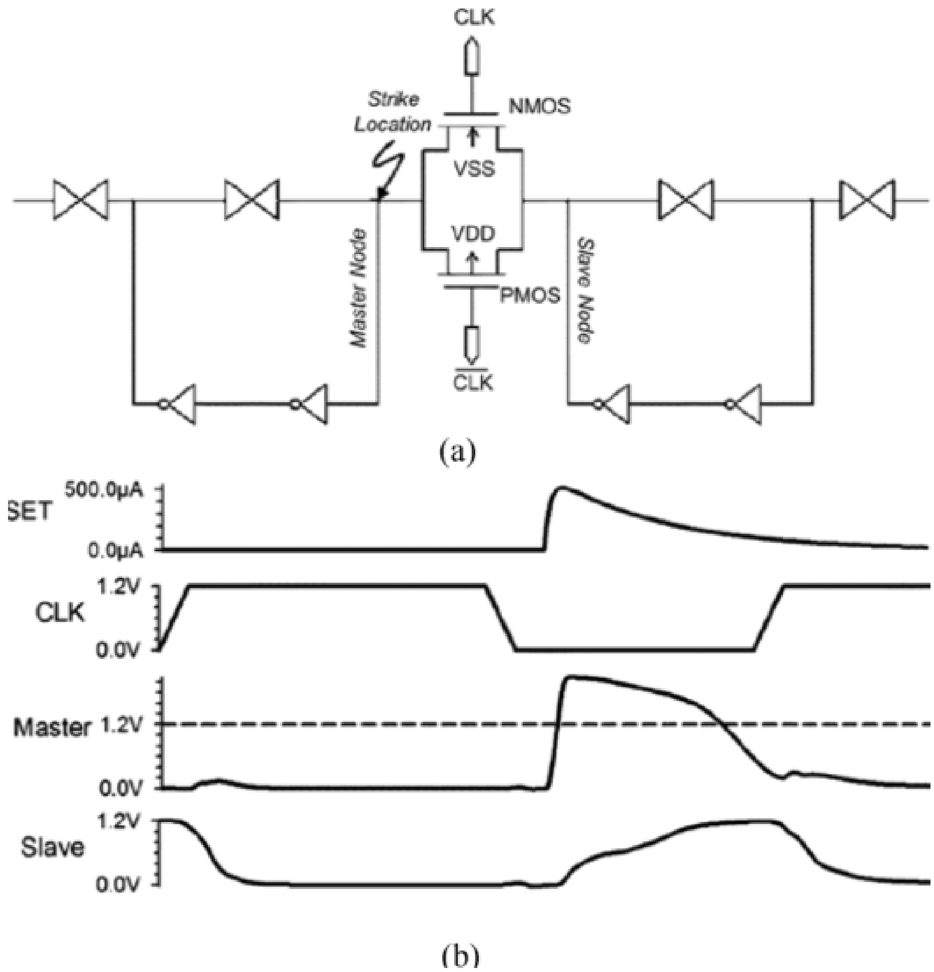


Figure 2.4: Single-event transient capture in a master–slave D flip-flop. *(a)* Circuit schematic showing the strike location at the internal transmission gate, with the VSS-connected NMOS transistor as the struck node and the CLK and $\overline{\text{CLK}}$ signals controlling the sampling windows. *(b)* Simulated waveforms: the SET current pulse (top), clock (second), master node voltage (third, with dashed threshold line), and slave node output (bottom). The upset is captured when the master node exceeds the logic threshold during the active clock window and is subsequently propagated to the slave output.

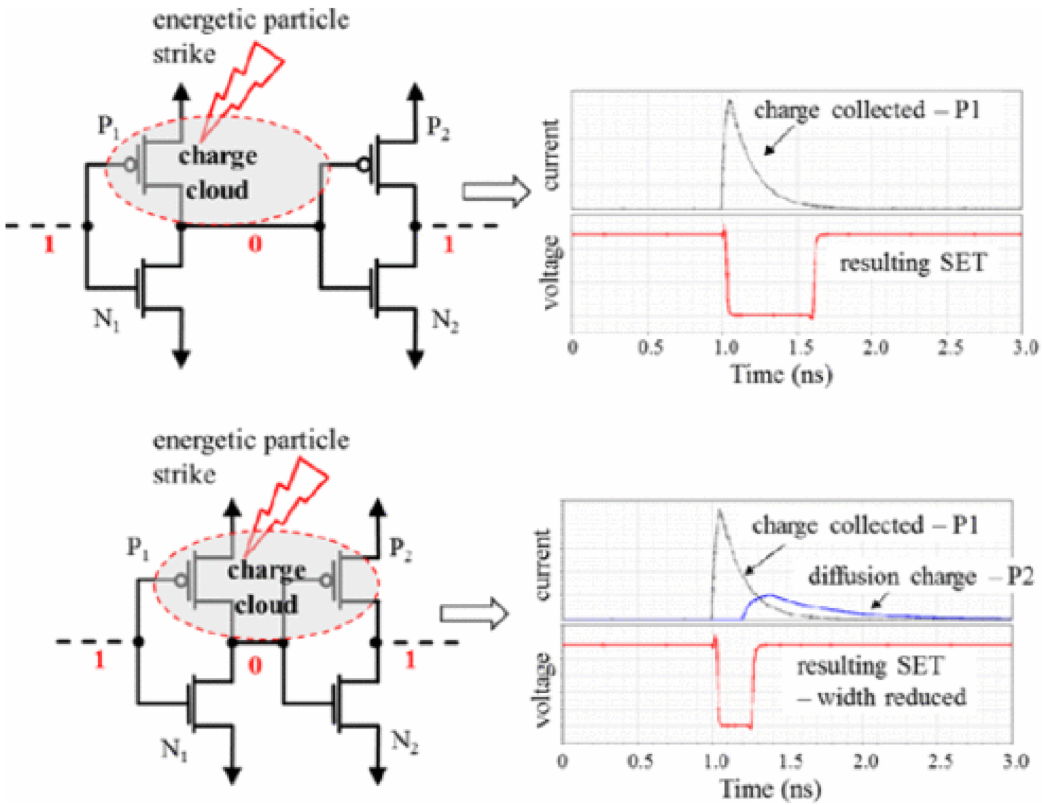


Figure 2.5: Effect of inter-transistor charge sharing on SET pulse width. *(Upper row)* Without charge sharing, the charge collected at transistor P_1 is unconstrained, yielding a full-width voltage transient at the logic output. *(Lower row)* When the adjacent transistor P_2 intercepts a fraction of the diffusion charge cloud, the net charge delivered to P_1 is reduced, resulting in a narrower SET pulse that is less likely to meet the setup-time requirement of a downstream register.

power-cycling circuitry is a standard component of irradiation test setups for CMOS devices.

2.2.4 Single-event functional interrupt

A single-event functional interrupt (SEFI) is a condition in which a particle strike causes a device to enter an unintended operating mode or to cease normal operation [53]. Unlike a simple data upset, a SEFI typically affects control logic, configuration registers, or state machines that govern global device behavior. Examples include the corruption of a clock management unit, the assertion of an internal reset, or the entry into a test or debug mode.

SEFIs are of particular concern in complex devices such as FPGAs, microprocessors, and high-density memories where internal control logic is not directly visible to the user. Recovery from a SEFI generally requires a device reset or reconfiguration, which disrupts operation and can result in data loss. SEFIs can be difficult to distinguish from other failure modes without dedicated monitoring of control and status signals.

2.2.5 Other destructive effects

Single-event burnout (SEB) and single-event gate rupture (SEGR) are destructive phenomena primarily associated with power devices. SEB occurs when a particle strike triggers a second breakdown mechanism in a power transistor, resulting in localized thermal runaway. SEGR occurs when charge deposited near a gate oxide causes the electric field across the oxide to exceed its breakdown strength [52]. While these effects are not the primary focus of this dissertation, they are noted for completeness and because power supply components in FPGA systems may be vulnerable [31].

The physical mechanism of SEGR is illustrated in Figure 2.6. When a heavy ion traverses the gate oxide of a power MOSFET under a large drain-source bias and a negative gate voltage, the ion track provides a transient conductive column through the dielectric. Holes accumulate at the inverted Si–SiO_2 interface while electrons are rapidly swept to the n^+ drain contact. The resulting local enhancement of the electric field across the oxide can exceed the intrinsic breakdown strength, causing permanent dielectric rupture.

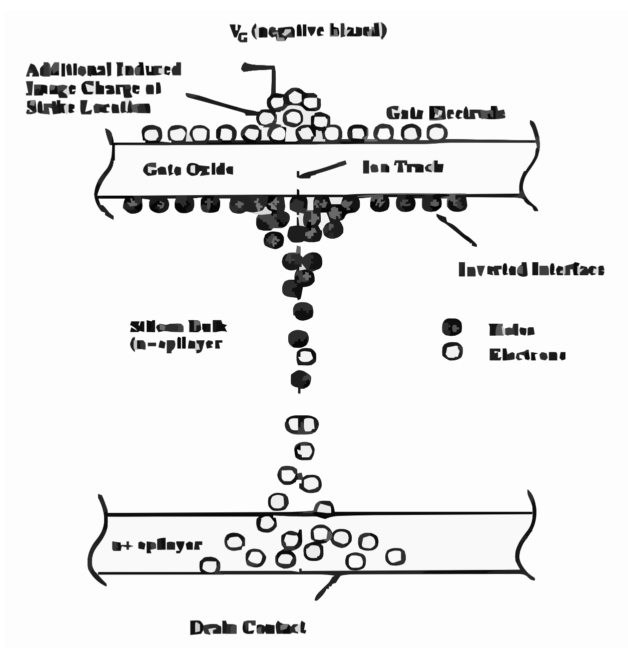
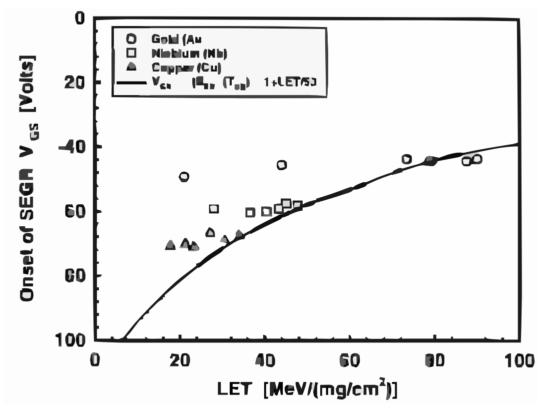


Figure 2.6: Cross-sectional diagram of the single-event gate rupture (SEGR) mechanism in a power MOSFET. A negatively biased gate ($V_G < 0$) and a large drain-source voltage establish a high electric field across the gate oxide. When a heavy ion traverses the structure, electron–hole pairs are generated along the ion track; holes migrate to and accumulate at the inverted Si–SiO_2 interface while electrons drift toward the n^+ drain contact. The localized field enhancement at the interface can drive the oxide field beyond its breakdown threshold, resulting in permanent device failure.

The onset of SEGR is a strong function of both the ion LET and the applied drain-source bias. Figure 2.7 shows measured SEGR cross-sections as a function of V_{DS} for four ion species spanning a range of LET values. Figure 2.8 quantifies the onset condition by plotting the critical gate voltage at which SEGR occurs as a function of ion LET for several species. The empirical model $V_{\mathrm{GS}} = (E_{\mathrm{BD}})(T_{\mathrm{OX}})/(1 + \mathrm{LET}/53)$ captures the monotonic reduction in breakdown threshold with increasing LET [52].



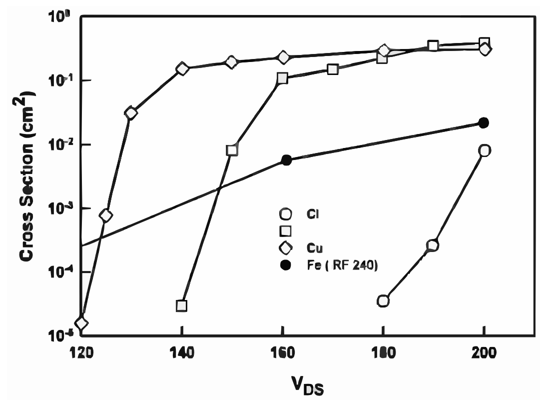


Figure 2.7: Measured SEGR cross-section as a function of drain-source voltage V_{DS} for chlorine (Cl), iodine (I), copper (Cu), and iron (Fe) ions. Higher-LET ions exhibit lower onset voltages and higher saturation cross-sections.

Figure 2.8: Onset gate voltage for SEGR as a function of ion LET for gold (Au), niobium (Nb), and copper (Cu) ions. The solid curve is the empirical Sexton model. The threshold gate voltage decreases with increasing LET, reflecting the progressive reduction in the additional field increment needed to reach breakdown.

2.3 Architecture of field-programmable gate arrays

Understanding how radiation effects manifest in FPGAs requires a detailed appreciation of the device architecture, because the physical location and function of an upset bit determine its consequences for the implemented circuit. This section describes the major architectural elements of SRAM-based FPGAs, which represent the most widely used and radiation-sensitive family. The foundational concepts of FPGA architecture—programmable logic, routing fabrics, and configuration mechanisms—are described by Rose et al. [49], with a comprehensive survey of architectural evolution and trade-offs provided by Kuon, Tessier, and Rose [26].

2.3.1 Overall device organization

An SRAM-based FPGA consists of a regular array of configurable logic blocks (CLBs) embedded in a mesh of programmable routing resources, surrounded by a ring of programmable input/output blocks (IOBs). Modern devices augment this fabric with embedded hard macros: block random-access memory (block RAM), digital signal processing (DSP) slices, clock management tiles (PLLs and delay-locked loops), high-speed serial transceivers, and in some families, embedded processor cores [23].

The entire functionality of the device is determined by a configuration bitstream that is loaded into an array of SRAM cells at power-up. This bitstream programs the truth tables of lookup tables, the states of multiplexers in the routing network, the initial values of user flip-flops, and the operating modes of hard macros. The total number of configuration bits ranges from a few million in small devices to over one hundred million in large modern FPGAs. Because these configuration SRAM cells use the same technology as the user-accessible storage, they are subject to the same radiation-induced upset mechanisms.

2.3.2 Configurable logic blocks and lookup tables

The CLB is the fundamental computational unit of the FPGA fabric. In a typical architecture (using Xilinx terminology as a representative example), each CLB contains two or more slices, and each slice contains a set of lookup tables (LUTs), flip-flops, carry logic, and local multiplexers.

A K-input LUT implements any Boolean function of up to K variables by storing the complete truth table in 2^K SRAM cells and using the input signals as select lines for a 2^K-to-1 multiplexer tree. For a 6-input LUT, which is common in modern architectures, this requires 64 SRAM configuration cells per LUT. The stored values define the logic function: changing any one of these configuration bits alters the truth table and therefore changes the implemented function for at least one input combination.

Each slice also contains flip-flops (typically D-type) that can optionally register the LUT output. These flip-flops are part of the user design and hold the sequential state of the implemented circuit. Additional multiplexers within the slice allow the LUT to be used as a small distributed RAM or as a shift register in some architectures.

The carry chain provides a dedicated fast-propagation path for arithmetic operations. It bypasses the general routing network and enables efficient implementation of adders, counters, and comparators with predictable timing.

2.3.3 Programmable routing and interconnect

The routing network consumes the majority of the configuration bitstream and the silicon area in an SRAM-based FPGA. Interconnect resources are organized hierarchically: local connections within a CLB, short routing segments spanning a few CLB tiles, and long lines spanning entire rows or columns.

Routing is accomplished through programmable interconnect points (PIPs), each controlled by one or more SRAM configuration cells. A PIP is typically a pass-transistor or a buffered multiplexer whose control input is driven by a configuration SRAM cell. When the SRAM cell stores a logic '1', the PIP connects two routing segments; when it stores a logic '0', the segments are isolated.

The implication for radiation sensitivity is significant. Because the routing network is controlled by configuration SRAM cells, an upset in a routing PIP can connect signals that should be isolated, disconnect signals that should be connected, or short two driven signals together. These routing errors can produce failure modes that are qualitatively different from simple data corruption: a bridged net may create a contention condition, and an open net may leave a downstream input floating or driven to an unintended value [23, 32].

2.3.4 Block RAM and DSP slices

Block RAM provides dedicated high-density storage implemented as true dual-port SRAM. Block RAM contents are part of the user data space and are subject to SEU in the same manner as any SRAM. Unlike configuration memory, block RAM upsets affect stored data directly without altering the implemented circuit. Error correcting codes can be applied to block RAM contents either through built-in ECC logic or through user-instantiated protection.

DSP slices are hard-wired multiply-accumulate units optimized for signal processing. They contain dedicated registers and arithmetic logic with fixed datapaths. Upsets in DSP slice registers affect computed values, while upsets in the configuration bits that control the DSP operating mode can alter its function.

2.3.5 Clock distribution and management

FPGAs include dedicated clock distribution networks with low skew and high fan-out, driven by clock management tiles that contain phase-locked loops (PLLs) or mixed-mode clock managers. These elements generate internal clocks with controlled frequency and phase from external reference signals.

Radiation-induced upsets in clock management configuration bits can alter the frequency multiplication or division ratios, change phase offsets, or disable the PLL entirely. Because the clock is a global resource, a single upset affecting clock generation can disrupt the timing of the entire design. SEFIs involving clock management are among the most consequential failure modes in irradiated FPGAs [1].

2.3.6 Configuration memory organization

The configuration memory of an SRAM-based FPGA is organized as a two-dimensional array of frames. A frame is the smallest addressable unit for readback and partial reconfiguration. Each frame typically spans the full height of a clock region and contains configuration bits for logic, routing, and embedded resources within that column. The total configuration space is accessed through a dedicated internal configuration access port (ICAP) or through external interfaces such as JTAG [60].

This frame-based organization has direct implications for scrubbing and readback strategies. Configuration verification can be performed by reading back frames and comparing them against a known-good reference. Correction is performed by rewriting the corrupted frame. The granularity of this process—one frame at a time—determines the minimum latency from detection to repair and influences the design of scrubbing controllers [18, 58].

2.4 Radiation effects specific to FPGAs

The architecture described in Section 2.3 creates a distinctive landscape of radiation vulnerabilities that differs from that of an ASIC or a microprocessor. Wirthlin provides an overview of reliability challenges in FPGA-based systems for space and high-energy physics, emphasizing the interplay between architecture and radiation response [59]. Quinn et al. review how Xilinx FPGA architectural features have evolved from the Virtex through Virtex-5 families with respect to radiation reliability, including the growing importance of multi-bit upsets [45]. This section categorizes FPGA-specific radiation effects by the resource affected.

2.4.1 Configuration memory upsets

Upsets in the configuration RAM (CRAM) are the dominant failure mode in SRAM-based FPGAs [38, 23]. A single configuration bit flip can alter a LUT truth table entry, change a routing multiplexer selection, modify a block RAM initialization value, or reconfigure a hard macro operating mode. Because CRAM upsets change the *circuit* rather than the *data*, their effects are persistent: the fault remains until the corrupted frame is scrubbed or the device is fully reconfigured.

The functional impact of a CRAM upset depends on whether the affected bit is *essential*—meaning it contributes to the implemented circuit—or *non-essential*—meaning it corresponds to unused resources. The fraction of essential bits varies with design utilization and is typically between 10% and 40% of the total configuration space [24]. Among essential bits, a further distinction can be drawn between bits whose corruption produces an immediately observable error and bits whose corruption affects a rarely exercised path or a masked input combination.

2.4.2 User logic upsets

Upsets in flip-flops and other user storage elements (distributed RAM, shift registers) affect the sequential state of the implemented design. These upsets are transient in the sense that the corrupted state will be overwritten by normal circuit operation on subsequent clock cycles, provided the design continues to function. However, if the upset occurs in a feedback path or a control register, the erroneous state may persist indefinitely or trigger a cascade of errors through the design's state machine [32].

User logic upsets differ from CRAM upsets in their observability and recovery characteristics. A data register upset may be detected through checksums or redundancy comparisons, and the register will be refreshed on the next valid write. A state machine upset, by contrast, may drive the system into an unrecoverable state that requires external intervention.

2.4.3 Single-event transients in FPGA combinational paths

SETs in FPGA logic paths are mediated by the LUT-based architecture. A transient glitch at a LUT input or within the multiplexer tree of a LUT can propagate to the LUT output and be captured by a downstream flip-flop. The probability of capture depends on the pulse width relative to the setup and hold windows of the capturing register and on whether the affected logic path is currently active.

The programmable routing in an FPGA can also introduce SET-like behavior when charge is deposited on an interconnect segment or at a routing switch. Because routing segments can have significant capacitance and are driven by pass-transistor switches, a transient voltage perturbation on a routing line may propagate along the net and appear at multiple destinations [12].

2.4.4 Single-event functional interrupts in FPGAs

FPGAs contain internal control and housekeeping logic that manages configuration, clock generation, power-on sequencing, and in some cases built-in self-test functions. Upsets affecting this control logic can cause SEFIs that disrupt global device operation. Examples include corruption of the global set/reset network, inadvertent entry into configuration mode, loss of PLL lock, and disruption of the JTAG state machine [1].

SEFIs are particularly insidious because they may not be detectable by monitoring primary design outputs alone. A PLL that loses lock may cause the design to stop clocking without producing an explicit error signal on any user-visible pin. Detection of SEFIs typically requires dedicated health monitoring of device status registers and supply currents, which must be implemented external to the FPGA or in a radiation-hardened supervisory controller.

2.5 Mitigation and hardening strategies

Mitigation strategies for FPGA radiation effects span a hierarchy from the transistor level to the system level. The choice of strategy involves trade-offs among resource overhead, performance impact, recovery latency, and residual vulnerability. Figure 2.9 presents a taxonomy of the principal fault-tolerance mechanisms applicable to radiation-hardened digital systems; subsequent subsections address the most relevant categories in detail.

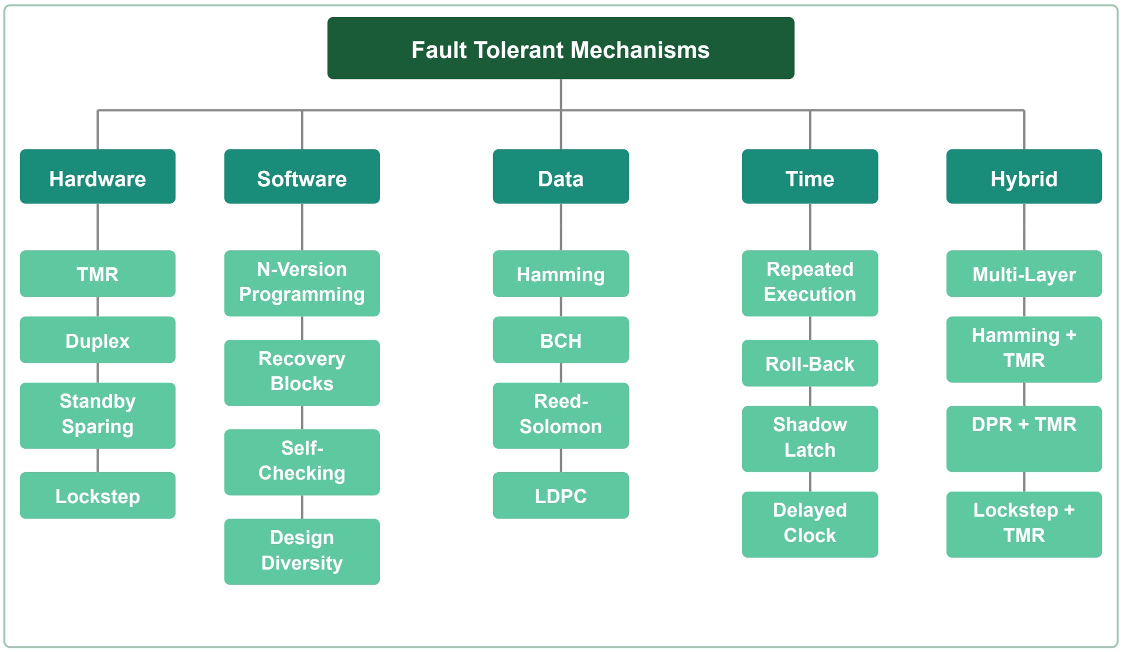


Figure 2.9: Taxonomy of fault-tolerant design mechanisms for radiation-hardened digital systems. Strategies are organized into five categories: *hardware* redundancy (TMR, duplex, standby sparing, lockstep); *software* techniques (N-version programming, recovery blocks, self-checking, design diversity); *data-encoding* schemes (Hamming, BCH, Reed–Solomon, LDPC); *time-based* methods (repeated execution, roll-back, shadow latch, delayed clock); and *hybrid* combinations that layer multiple mechanisms to achieve higher coverage at reduced overhead relative to pure hardware triplication.

2.5.1 Radiation-hardened FPGA technologies

Radiation-hardened by design (RHBD) and radiation-hardened by process (RHBP) FPGAs incorporate modifications at the transistor and cell level to increase critical charge and reduce charge collection efficiency. Examples include the use of silicon-on-insulator (SOI) substrates, guard rings around SRAM cells, and resistive-feedback latches. These devices offer inherently lower SEU cross-sections but are more expensive, available in older technology nodes, and subject to longer procurement cycles than commercial devices [23, 34]. The trade-offs between radiation-hardened and commercial-grade devices are increasingly important as the space industry adopts COTS components for cost and performance reasons [42]; a recent systematic review by Dijks et al. maps the evolving engineering and testing approaches for radiation hardness assurance in COTS space avionics, identifying modular design, hybrid fault tolerance, and agile development processes as central themes [9].

Non-volatile configuration technologies, such as flash-based and antifuse-based FPGAs, eliminate the configuration SRAM vulnerability entirely. Flash-based devices (e.g., the Microsemi RTG4 and IGLOO2 families) store configuration in flash cells that are substantially more resistant to single-event upsets than SRAM cells. The trade-off is that these families offer lower logic density and fewer features than leading-edge SRAM-based devices.

2.5.2 Triple modular redundancy

Triple modular redundancy (TMR) replicates a logic function three times and passes the outputs through a majority voter [30]. If any single replica produces an incorrect output due to an upset, the voter output remains correct. TMR is widely used in FPGA designs for radiation environments and can be applied at various granularities: fine-grain TMR replicates individual flip-flops or LUTs, while coarse-grain TMR replicates entire functional blocks.

TMR approximately triples the resource utilization and increases power consumption. The voter itself represents a potential single point of failure unless it is also triplicated [56, 22]. Morgan et al. compare TMR with alternative fault-tolerant techniques for FPGAs, including duplication with comparison and partial TMR, finding that TMR provides the best error coverage at the cost of the greatest area overhead [33]. Furthermore, TMR protects only against single faults within a voting domain; if two replicas are simultaneously corrupted, the voter will select an incorrect value. Quinn et al. demonstrated that even two-bit MCUs in the routing network can produce domain crossing errors that defeat TMR on a single FPGA device [46]. Cannon and Wirthlin subsequently developed placement strategies to reduce such common-mode failures in TMR designs on SRAM FPGAs [7]. These limitations motivate the use of TMR in combination with configuration scrubbing to ensure that accumulated configuration upsets are corrected before multiple replicas are compromised [43].

Figure 2.10 illustrates the decision logic of a TMR majority voter. Three independent replicas feed the voter, which implements 2-of-3 agreement logic. When all three outputs are identical, the agreed value is passed to downstream logic with no error indication. When exactly two replicas agree, the majority value is selected and the divergent module is flagged. When no two replicas agree—a triple-module fault—the output is undefined and must be handled at the system level, for example by triggering a full reconfiguration or asserting a safe-state output.

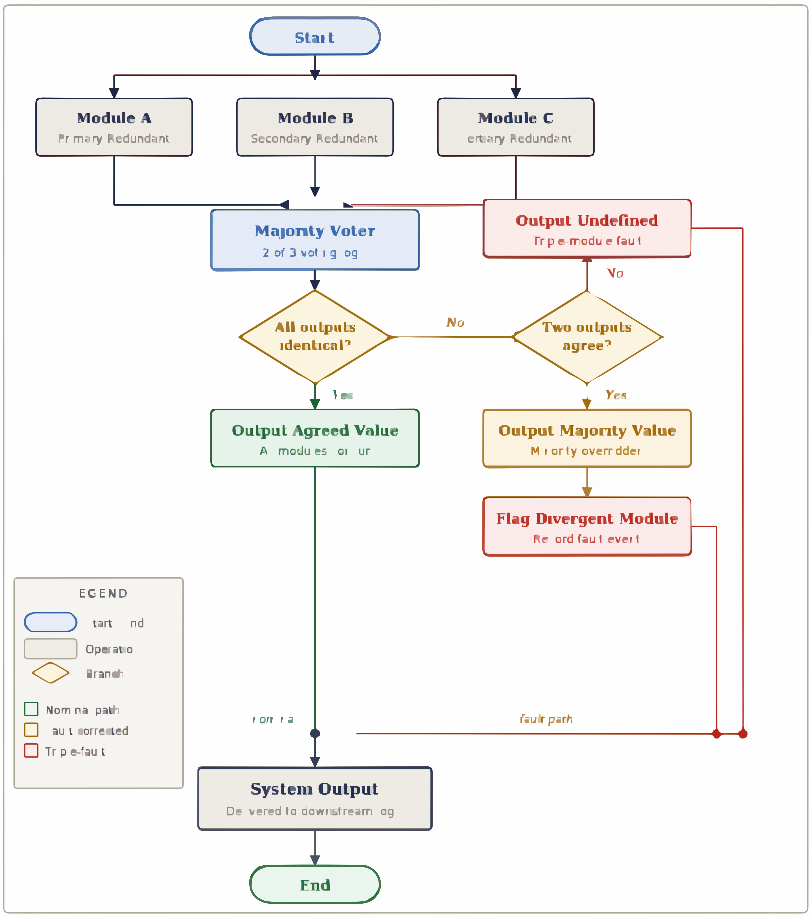


Figure 2.10: Decision flowchart for a triple modular redundancy (TMR) majority voter implementing 2-of-3 voting logic. Three independent replicas (Modules A, B, C) feed the voter. When all outputs agree, the result passes through the nominal path. When two outputs agree, the majority value is selected and the minority module is flagged as divergent (fault-corrected path). When no two outputs agree, a triple-module fault is declared and the output is undefined. This last condition motivates combining TMR with configuration scrubbing so that accumulated upsets are corrected before a second replica is corrupted.

2.5.3 Configuration scrubbing

Configuration scrubbing is the periodic process of reading back the configuration memory, detecting errors, and writing correct values to restore the intended circuit [18]. Scrubbing addresses the persistent nature of CRAM upsets by limiting the time that a corrupted configuration bit remains in the device.

Scrubbing can be performed externally (by a separate controller reading and writing the configuration port) or internally (by logic implemented within the FPGA itself, using the ICAP) [5, 58]. Herrera-Alzu and López-Vallejo present a systematic classification of scrubber design techniques for Xilinx Virtex FPGAs, including considerations for radiation-tolerant and space-qualified families [19]. Internal scrubbing offers lower latency but consumes FPGA resources and may itself be vulnerable to upsets; early work by Graham et al. identified half-latches in the Xilinx routing network as a specific vulnerability that must be addressed by scrubbing strategies [17]. Hybrid approaches use a combination of internal error detection (via CRC or ECC on configuration frames) and external correction.

The scrub rate determines the mean time to repair (MTTR) for configuration upsets. A complete scrub of a modern FPGA may take hundreds of milliseconds to several seconds depending on the configuration interface bandwidth and the device size [60]. The scrub interval must be chosen so that the probability of accumulating multiple upsets in the same TMR domain before scrubbing corrects them is acceptably low.

2.5.4 Error detection codes for block RAM and user memory

Block RAM and other on-chip memories can be protected using error correcting codes. Single-error correction, double-error detection (SECDED) Hamming codes are commonly employed because they add moderate overhead and can correct single-bit upsets in a codeword. For memories where MCUs within a single codeword are possible, interleaving of codeword bits across physical locations can maintain single-bit correction capability [23].

2.5.5 Watchdog timers and system-level health monitoring

Circuit-level mitigations address specific fault mechanisms but cannot guarantee correct system behavior in the presence of unanticipated failure modes. System-level health monitoring provides a complementary defense by observing the external behavior of the system and triggering recovery actions when anomalies are detected.

Watchdog timers verify that the system continues to execute expected sequences within bounded time. A watchdog that is not periodically serviced by the monitored logic triggers a recovery action, typically a partial or full reset. More sophisticated health monitors can track application-specific indicators such as control loop period, telemetry frame rate, or checksum validity.

The design of health monitoring for irradiation experiments must balance sensitivity (detecting real faults) against specificity (avoiding false alarms from transient perturbations that the system would naturally recover from). This balance is application-dependent and is a recurring theme in the experimental work described in later chapters.

2.6 Prior work in FPGA irradiation testing

Experimental evaluation of FPGA radiation sensitivity has evolved from component-level cross-section measurements toward increasingly system-oriented methodologies. Figure 2.11 summarizes the principal testing paradigms currently employed, organized by the nature of the test stimulus and the observability it provides. This section reviews each paradigm and identifies the limitations that motivate the present work.

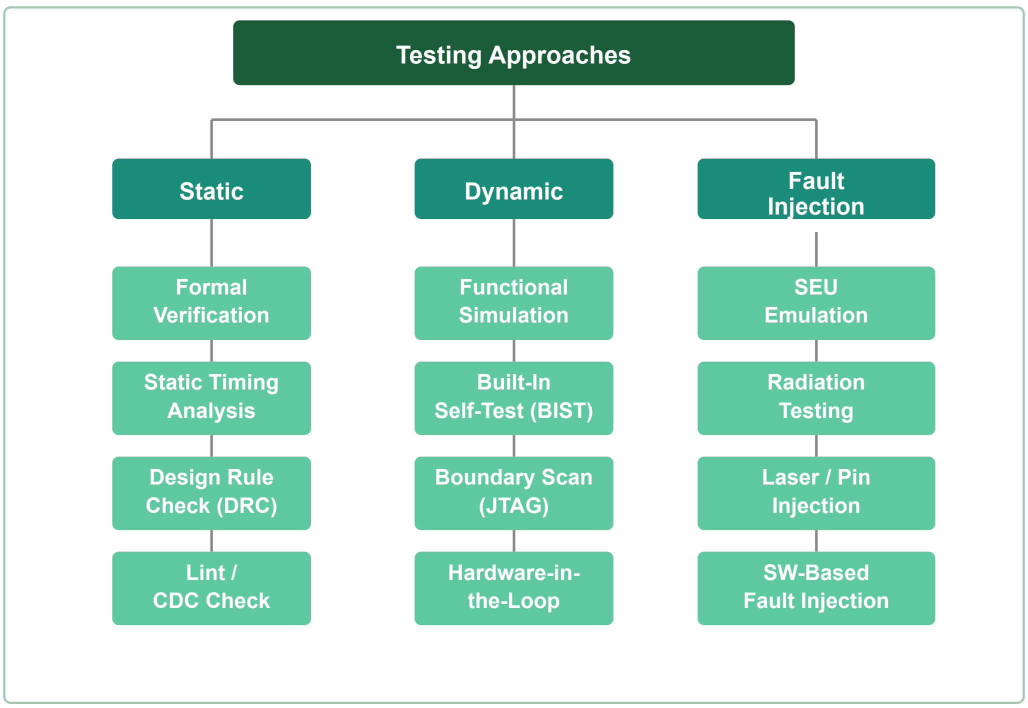


Figure 2.11: Taxonomy of testing approaches for radiation-hardened digital systems. *Static* methods (formal verification, static timing analysis, design rule checking, lint/CDC) provide pre-silicon assurance of structural and temporal correctness. *Dynamic* methods (functional simulation, built-in self-test, boundary scan via JTAG, hardware-in-the-loop) evaluate behavior during operation. *Fault injection* techniques (SEU emulation, radiation beam testing, laser and pin injection, software-based fault injection) characterize radiation response directly and can expose failure modes inaccessible to static or dynamic methods alone.

2.6.1 Static cross-section measurements

Early FPGA radiation testing focused on measuring the static cross-section of configuration memory by loading a known bitstream, exposing the device to a particle beam, and reading back the configuration to count altered bits [38]. This approach yields a well-defined cross-section as a function of LET or particle energy and provides the fundamental data needed for rate predictions using established methodologies such as those reviewed by Petersen et al. [41]. The JEDEC JESD89A standard formalizes the measurement and reporting of soft errors induced by alpha particles and cosmic rays [21]. Lesea et al. extended static characterization by measuring atmospheric soft error rates across multiple FPGA technology generations in the Rosetta experiment [29]. However, static readback testing does not capture user logic upsets, SETs, or system-level failure modes because the design is not actively operating during readback.

2.6.2 Dynamic application-based testing

Dynamic testing evaluates a functioning FPGA design under irradiation by monitoring its outputs for errors while accumulating fluence [24, 2]. Test circuits range from simple shift registers and counters to application-representative benchmarks and full systems. Dynamic testing captures the combined effects of CRAM upsets, user logic upsets, and SETs as they manifest in observable behavior.

The challenge of dynamic testing lies in achieving adequate observability: the relationship between an observed output error and the underlying upset mechanism is often ambiguous. If the only monitoring is comparison of primary outputs against a golden reference, subtle internal faults that do not propagate to outputs within the observation window are missed. Wirthlin and colleagues demonstrated dynamic SEU sensitivity estimation using on-chip error detection logic operating in lockstep with the design under test [61, 28]. This approach improves observability by detecting internal state deviations before they propagate to outputs but requires additional FPGA resources and careful design to avoid common-mode vulnerabilities.

2.6.3 Fault injection as a complement to beam testing

Beam testing is expensive and access-limited, motivating the use of fault injection to supplement irradiation experiments. Configuration bit faults can be emulated by selectively flipping bits in the configuration memory using the ICAP or JTAG interface [56]. Sonza Reorda, Sterpone, and Violante demonstrated that single upsets in the configuration memory can produce multiple errors in the implemented circuit, motivating fault injection campaigns that go beyond simple single-bit flips [55]. Fault injection allows systematic exploration of the configuration space and identification of sensitive bits without a particle beam.

While fault injection is valuable for design validation and sensitivity analysis, it has limitations as a substitute for beam testing. Injection campaigns typically flip one bit at a time and observe the result, whereas real radiation produces complex temporal correlations, multi-bit upsets, and transient effects that are difficult to replicate through configuration manipulation alone.

2.6.4 Limitations of existing approaches and motivation for this work

The prior work summarized above has established the fundamental radiation sensitivities of FPGA technologies and demonstrated a range of mitigation techniques.

However, several gaps remain in the transition from component-level characterization to system-level evaluation under operational conditions. Quinn highlighted the inherent difficulty of testing complex systems, noting that as device integration increases, the number of possible failure modes grows faster than the ability to test for them exhaustively [44]. Coronetti et al. have advanced the case for radiation hardness assurance through system-level testing, establishing facility requirements, test methodologies, and data exploitation strategies for mixed-field environments [8]. Stirk et al. recently compared bare-metal and OS-based neutron radiation testing approaches for a complex SoC, demonstrating the complementary value of both methods for capturing different failure modes [57]. In-orbit anomaly data compiled by Ecoffet further underscore the need for testing methods that capture system-level failure modes not revealed by component-level characterization [13].

First, most published FPGA irradiation experiments use test structures (shift registers, multipliers, counters) that, while useful for cross-section measurement and benchmark comparison, do not exercise the closed-loop interactions characteristic of real instrumentation and control systems. The fault masking and fault amplification behavior of feedback control loops is underexplored in the FPGA radiation testing literature.

Second, observability in dynamic testing is often limited to primary output comparison, which provides a binary correct/incorrect classification but little insight into internal fault propagation, timing, or recovery dynamics. On-chip instrumentation that provides continuous visibility into fault activity without overwhelming telemetry bandwidth is needed.

Third, experimental automation and data management practices vary widely across published studies, making it difficult to reproduce experiments or to aggregate results across test campaigns. A systematic workflow that encompasses run segmentation, metadata capture, telemetry parsing, and post-processing is necessary to support quantitative analysis at the system level.

These gaps motivate the experimental infrastructure and methodology described in Chapter 3 and the closed-loop workload presented in Chapter 4. The need for system-level validation approaches that bridge the divide between component-level characterization and operational assurance is a recurring theme in recent reviews of radiation hardness assurance practice for COTS-based systems [9, 51].

Chapter 3: How to Actually Have a Cow

To actually have a cow, one must read [54] and look at the Cow Car in Figure 3.1. The table shown in Table 3.1 was generated using the data in Appendix A and further justifies this research.

3.1 Captions of Figures and Tables in LaTeX

In both figures and tables, you will want to have some sort of caption describing your work. These captions are generated using the command `\caption{`*text*`}` where *text* is your table/figure description. When this command is placed in a figure (see Section 3.2), it will generate the output "Figure *M.N*: *text*" where *M* is the chapter number and *N* is the sequence number of this figure (starting with Figure 1.1) and *text* is the text you have specified. If the `\caption` command is used in a table environment (see Section 3.3), you will get "Table *M.N*: *text*" where *M* is the chapter

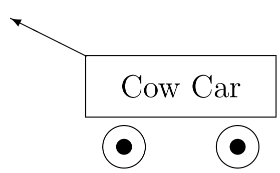


Figure 3.1: An example figure.

Summary of Cows Had	
Year	Number of Cows
1990	57
1991	80
1992	199

Table 3.1: An example table.

Summary of Cows Had	
Year	Number of Cows
1990	57
1991	80
1992	199

Table 3.2: Same example table but with an unnecessarily long name so as to cause it to go to multiple lines in the List of Figures.

number and N is the sequence number of this table (starting with Table 1.) and *text* is the text you have specified.

One important thesis style note (see [50]) is that the captions in figures *and* tables should come *after* the figure/table. This is different than in previous format specifications from the Graduate School.

3.2 Figures in LaTeX

Figures are included in LaTeX using the `figure` environment:

```
\begin{figure}
stuff for your figure
\caption{Description of your figure}
\end{figure}
```

Normally it is a pain to draw figures using TEX/LATEX commands. The picture shown in Figure 3.1 was produced using the `picture` environment.

3.2.1 "Pasting" Existing Figures in LATEX

Occasionally, you will want/need to include figures and pictures generated by other people. To do this, you will basically want to have LATEX leave some blank space for you and then (after the document is printed), use a copy machine to "paste" your figure into the document. This is easily done by using the `\vspace*` command. This command will leave a specified amount of vertical space in your document. For example, to leave 3 inches of vertical space, you would type

```
\vspace*{3in}
```

If you do make use of someone else's figures, you *must* have written permission from the creator of the figure to use it in your dissertation. *Be sure you have all required permissions before you attempt to turn in your thesis to the graduate school!!* Permission via email is usually adequate.

3.2.2 Postscript figures in LATEX

> Note: This section is written assuming you are using a *PostScript* printer such as those available in CIS and EE. If you are using LATEX 2$_\varepsilon$ in another department, please check with the system staff there to make sure this section applies to you.

A lot of people prefer to generate their pictures using WYSIWYG[1] programs like *idraw*, *xfig*, *MacDraw*, *FrameMaker*, etc. These result in a *PostScript* file which contains the picture. Figures in postscript can also be produced as the output of a graphics program, or as the dump of a window in a windowing system. These can be

[1] *W*hat *Y*ou *S*ee *I*s *W*hat *Y*ou *G*et.

included in your document using the LaTeX 2_ε package `epsfig`. In the portion where the "*stuff for your figure*" goes, use the `\epsfig` command to include the postscript file. When the document is finally printed on a postscript printer using the `dvips` program, the postscript file will be printed out as part of the document.

The Recommended Method

The recommended method is as follows. It uses the package `epsfig`, and assumes that the postscript file conforms to the *Encapsulated Postscript Script* standards. This calculates the amount of space the figure will take up, and automatically reserves space within the document for the postscript figure. The command `\epsfig` can be used as follows:

```
\epsfig{file=psfilename,width=desiredwidth}
```

An example command would be

```
\epsfig{file=dsmfig.ps,width=4in}
```

where `dsmfig.ps` is the name of the postscript file.

3.3 Tables in LaTeX

Tables are created using the `table` environment:

`\begin{table}`
stuff for your table (using tabular?)
`\caption{`Description of your table`}`
`\end{table}`

Most tables are created using the `tabular` environment. Details on using the `tabular` environment can be found in Lamport's LaTeX 2_ε book [27]. An example table is shown in Table 3.1.

Of course, you can also use the method of Section 3.2.1 to leave blank space and "paste" in a pre-made table after the document is printed as well.

Chapter 4: Contributions and Future Work

We have successfully shown in Chapter 3 how the problems of having a cow can be solved. A lot of work can still be done in this field so that better and bigger cows can be had. This is a matter which can be pursued by those weeny grad students who have just started their Ph.D.

In reality, this document has been prepared by Manas Mandal and Al Fencl to assist people in using the `osudissert96` class for LaTeX to generate masters theses and doctoral dissertations at the Ohio State University. The relevant files are available from the Department of Computer and Information Science or Electrical Engineering. Also, ask around your department; the files may have already migrated to a computer in your home department.

Keep in mind that this document *is not intended to teach you LaTeX* nor should it be used as such. Lamport's book [27] is a very good introduction to the wonderful world of LaTeX and it would be well worth your while to purchase that book!

It is our sincere hope that the use of this document and the related style files will help you in producing your thesis and/or dissertation. We wish you "Good Luck"!

Appendix A: The Data on Cows

This is the data that was used to produce the table in Table 3.1. In 1990, 57 people had cows. In 1991, 80 people had cows. In 1992, 199 people had cows.

Appendix B: Important commands defined in `osudissert96`

The following is a list of all commands available in `osudissert96`.

```
\author{First Middle Last}
\title{The title of the thesis}
\authordegrees{degree1, degree2}
\unit{Department of Whatever The Name Is}
\degree{Doctor of Philosophy}
\committee{Dissertation}
\advisorname{Name of advisor}  % Possible usage "Prof. Big Dude"
\member{Name of committee member}

\thesis  % this makes it a thesis rather than a dissertation,
         %  similar to the documentclass option [masters] or [ms].

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\begin{abstract}
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\dedication{This is dedicated to \ldots}

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\dateitem{Important Date}{ Why its important}

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% \majorfield{Your Major Field}
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%% Note:  If there were only one field of study, the following list
%%        would best be done using the following command:
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